



consider my current weights: Emotion Interpretation (Audio) 25% 30% High correlation with engagement states; validated in literature Video Saliency 22% 18% Strong visual attention indicator; proven engagement predictor Transcript Interpretation 20% 25% Content understanding crucial for engagement; semantic analysis reliability RMS + Audio Energy 15% 12% Reliable baseline measure; indicates vocal intensity and participation Spectral Features 12% 10% Complementary to RMS; captures vocal quality variations Pitch + Formants 6% 5% Useful but highly variable; cultural and individual differences

now take into consideration that I have engagement scores from a vocal versus non-vocal(background) and video motion/saliency data; vocal data has hubert SER, pitch/formants, rms_energy, spectral features, and transcript engagement scoring; non-vocal has rms_energy and spectral features engagement scoring, and video has a single engagement score based on motion, motion variance, saliency, and attention shift rate;

please tell me how to weigh all of these engagement scores into 1 engagement score based on good research

Based on my research into multimodal engagement detection and weighted fusion approaches, I can provide you with a research-backed framework for combining your engagement scores into a single comprehensive engagement metric.

Recommended Weighted Fusion Framework

Core Architecture: Dynamic Reliability-Based Late Fusion

Your optimal approach should use **late fusion with dynamic reliability-based weighting** rather than static weights. Research shows this consistently outperforms fixed weighting schemes.^{[1] [2] [3]}

Step 1: Reliability Assessment for Each Modality

First, calculate reliability scores for each engagement component:

Vocal Engagement (Your 63% total weight):

- **HuBERT SER reliability:** Based on confidence scores from the emotion recognition model and audio quality metrics (SNR, clarity)^{[4] [5]}
- **Pitch/Formants reliability:** Assess stability and consistency of extracted features; higher variability indicates lower reliability^{[6] [7]}
- **RMS Energy reliability:** Measure signal-to-noise ratio and dynamic range consistency^[6]
- **Spectral Features reliability:** Evaluate feature extraction confidence and spectral clarity^[4]
- **Transcript reliability:** Based on ASR confidence scores and semantic coherence^[8]

Non-vocal Background Engagement (Lower priority):

- **RMS Energy reliability:** Background noise consistency and signal quality
- **Spectral Features reliability:** Ambient sound clarity and consistency

Video Engagement:

- **Motion/Saliency reliability:** Based on tracking confidence, motion consistency, and visual quality metrics^{[9] [10]}

Step 2: Dynamic Weight Calculation Formula

Use this reliability-weighted fusion formula:^{[11] [2] [3]}

$$\text{Final_Engagement} = \frac{\sum(w_i \times s_i \times r_i)}{\sum(w_i \times r_i)}$$

Where:

- w_i = base weight from your current scheme
- s_i = engagement score from modality i
- r_i = reliability score for modality i (0-1 scale)

Step 3: Hierarchical Weighting Structure

Primary Level - Modal Categories:

- **Vocal Audio:** 65% (increased from your 63% due to higher reliability)
- **Video:** 25% (increased from 22% due to proven engagement correlation)^{[8] [9]}

- **Non-vocal Audio:** 10% (background/contextual support)

Secondary Level - Within Vocal Audio (65% total):

Based on research showing emotion interpretation and transcript analysis are most reliable: ^[1] ^[8]

- HuBERT SER: 35% ($0.35 \times 0.65 = 22.75\%$ of total)
- Transcript Interpretation: 30% ($0.30 \times 0.65 = 19.5\%$ of total)
- RMS Energy: 20% ($0.20 \times 0.65 = 13\%$ of total)
- Spectral Features: 10% ($0.10 \times 0.65 = 6.5\%$ of total)
- Pitch/Formants: 5% ($0.05 \times 0.65 = 3.25\%$ of total)

Step 4: Implementation Algorithm

```
def calculate_engagement_score(vocal_scores, nonvocal_scores, video_score,
                              vocal_reliability, nonvocal_reliability, video_reliability):

    # Base weights
    base_weights = {
        'vocal': 0.65,
        'video': 0.25,
        'nonvocal': 0.10
    }

    vocal_subweights = {
        'hubert_ser': 0.35,
        'transcript': 0.30,
        'rms': 0.20,
        'spectral': 0.10,
        'pitch': 0.05
    }

    # Calculate weighted vocal engagement
    vocal_engagement = sum(
        vocal_subweights[key] * vocal_scores[key] * vocal_reliability[key]
        for key in vocal_scores.keys()
    ) / sum(vocal_subweights[key] * vocal_reliability[key] for key in vocal_scores.keys())

    # Calculate final weighted engagement with reliability adjustment
    components = [
        (vocal_engagement, base_weights['vocal'], vocal_reliability['overall']),
        (video_score, base_weights['video'], video_reliability),
        (nonvocal_scores['combined'], base_weights['nonvocal'], nonvocal_reliability['overall'])
    ]

    weighted_sum = sum(score * weight * reliability for score, weight, reliability in components)
    total_weight = sum(weight * reliability for _, weight, reliability in components)

    return weighted_sum / total_weight
```

Step 5: Adaptive Confidence Thresholding

Implement confidence-based modality gating:^[12]

- If reliability score < 0.3: Reduce weight by 50%
- If reliability score < 0.1: Exclude modality from fusion
- Redistribute excluded weights proportionally to remaining modalities

Key Research-Backed Advantages

1. **Reliability-based weighting** prevents low-quality data from degrading performance^{[7] [12]}
2. **Late fusion** allows each modality to be optimized independently^{[2] [13]}
3. **Dynamic weighting** adapts to varying data quality in real-time^{[14] [3]}
4. **Hierarchical structure** captures both inter-modal and intra-modal relationships^[15]

This framework leverages proven multimodal fusion techniques while respecting your domain expertise in weighting the different engagement indicators. The reliability adjustment ensures robust performance even with varying audio/video quality conditions.

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